

LECTURE 5: HYBRID ELECTRIC VEHICLE

INTRODUCTION

Conventional vehicles with IC engines provide good performance and long operating range by utilizing the high-energy-density advantages of petroleum fuels. However, conventional IC engine vehicles have the disadvantages of poor fuel economy and environmental pollution. The main reasons for their poor fuel economy are

- 1) mismatch of engine fuel efficiency characteristics with the real operation requirement
- 2) dissipation of vehicle kinetic energy during braking, especially while operating in urban areas; and
- 3) low efficiency of hydraulic transmission in current automobiles in stop-and-go driving patterns.

Battery-powered EVs, on the other hand, possess some advantages over conventional IC engine vehicles, such as high-energy efficiency and zero environmental pollution. However, the performance, especially the operation range per battery charge, is far less competitive than IC engine vehicles, due to the much lower energy density of the batteries than that of gasoline. HEVs, which use two power sources (a primary power source and a secondary power source), have the advantages of both IC engine vehicles and EVs and overcome their disadvantages.

THE HYBRID ELECTRIC VEHICLE (HEV)

Basically, any vehicle power train is required to

- (1) develop sufficient power to meet the demands of vehicle performance,
- (2) carry sufficient energy on board to support the vehicle driving a sufficient range,
- (3) demonstrate high efficiency, and
- (4) emit few environmental pollutants.

Broadly, a vehicle may have more than one power train. Here, the power train is defined as the combination of the energy source and the energy converter or power source, such as the gasoline (or diesel)–heat engine system, the hydrogen–fuel cell electric motor system, the chemical battery–electric motor system, and so on. What exactly is an HEV? The definition available is so general that it anticipates future technologies of energy sources. The term *hybrid vehicle* refers to a vehicle with at least two sources of power. A *hybrid-electric vehicle* indicates

that *one source of power* is provided by an *electric motor* . The *other source of motive power* can come from a number of different technologies, but is typically provided by an *internal combustion engine* designed to run on either gasoline or diesel fuel. As proposed by Technical Committee (Electric Road Vehicles) of the International Electrotechnical Commission, *an HEV is a vehicle in which propulsion energy is available from two or more types of energy sources and at least one of them can deliver electrical energy* . Based on this general definition, there are many types of HEVs, such as:

- the gasoline ICE and battery
- diesel ICE and battery
- battery and FC
- battery and capacitor
- battery and flywheel
- battery and battery hybrids.

Most commonly, the propulsion force in HEV is provided by a combination of electric motor and an ICE. The electric motor is used to improve the energy efficiency (improves fuel consumption) and vehicular emissions while the ICE provides extended range capability.

Energy Use in Conventional Vehicles

In order to understand how a HEV may save energy, it is necessary first to examine how conventional vehicles use energy. The breakdown of energy use in a vehicle is as follows:

- In order to maintain movement, vehicles must produce power at the wheels to overcome:
 - aerodynamic drag (air friction on the body surfaces of the vehicle, coupled with pressure forces caused by the air flow)
 - rolling resistance (the resistive forces between tires and the road surface)
 - resistive gravity forces associated with climbing a grade
- Further, to accelerate, the vehicle must its inertia. Most of the energy expended in acceleration is then lost as heat in the brakes when the vehicle is brought to a stop.
- The vehicle must provide power for accessories such as heating fan, lights, power steering, and air conditioning.

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- Finally, a vehicle will need to be capable of delivering power for acceleration with very little delay when the driver depresses the accelerator, which may necessitate keeping the power source in a standby (energy-using) mode.

A conventional engine-driven vehicle uses its engine to translate fuel energy into shaft power, directing most of this power through the drivetrain to turn the wheels. Much of the heat generated by combustion cannot be used for work and is wasted, both because heat engines have theoretical efficiency limit. Moreover, it is impossible to reach the theoretical efficiency limit because:

- some heat is lost through cylinder walls before it can do work
- some fuel is burned at less than the highest possible pressure
- fuel is also burned while the engine is experiencing negative load (during braking) or when the vehicle is coasting or at a stop, with the engine idling.

Although part of engine losses would occur under any circumstances, part occurs because in conventional drivetrains, engines are sized to provide very high levels of peak power for the acceleration capability expected by consumers, about 10 times the power required to cruise at 100Km/h. However, the engines are operated at most times at a small fraction of peak power and at these operating points they are quite inefficient.

Having such a large engine also increases the amount of fuel needed to keep the engine operating when the vehicle is stopped or during braking or coasting, and increases losses due to the added weight of the engine, which increases rolling resistance and inertial losses. Even gradeability requirements (example: 55 mph up a 6.5% grade) require only about 60 or 70% of the power needed to accelerate from 0 to 100Km/h in under 12 seconds.

In **Figure 1** it is shown how fuel energy is translated into work at the wheels for a typical midsize vehicle in urban and highway driving. From **Figure 1** it can be observed that:

- At best, only 20% of the fuel energy reaches the wheels and is available to overcome the tractive forces, and this is on the highway when idling losses are at a minimum, braking loss is infrequent, and shifting is far less frequent.
- Braking and idling losses are extremely high in urban driving and even higher in more congested driving, e.g., within urban cores during rush hour. Braking loss, represents

46% of all tractive losses in urban driving. Idling losses represent about one sixth of the fuel energy on this cycle.

- Losses to aerodynamic drag, a fifth or less of tractive losses in urban driving, are more than half of the tractive losses during highway driving.

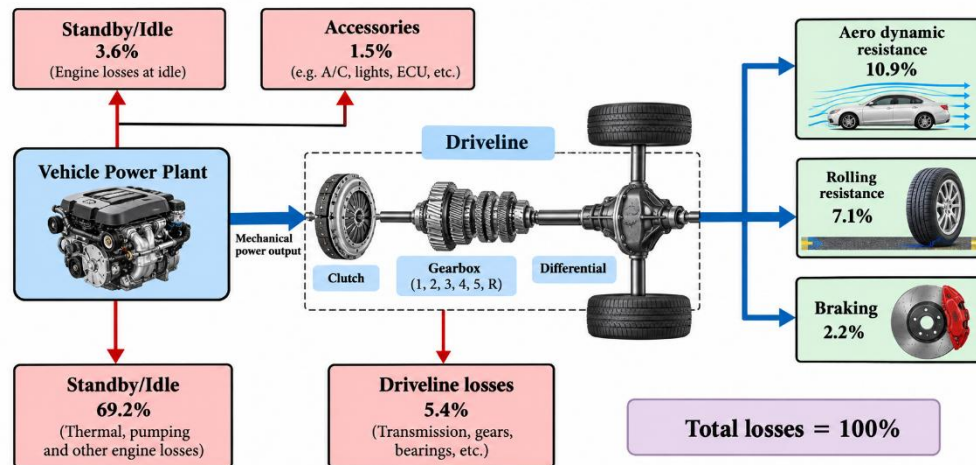


Figure 1: Translation of fuel energy into work in a vehicle

Energy Savings Potential of Hybrid Drivetrains

In terms of overall energy efficiency, the conceptual advantages of a hybrid over a conventional vehicle are:

- Regenerative braking.** A hybrid can capture some of the energy normally lost as heat to the mechanical brakes by using its electric drive motor(s) in generator mode to brake the vehicle
- More efficient operation of the ICE, including reduction of idle.** A hybrid can avoid some of the energy losses associated with engine operation at speed and load combinations where the engine is inefficient by using the energy storage device to either absorb part of the ICE's output or augment it or even substitute for it. This allows the ICE to operate only at speeds and loads where it is most efficient. When an HEV is stopped, rather than running the engine at idle, where it is extremely inefficient, the control system may either shut off the engine, with the storage device providing auxiliary power (for heating or cooling the vehicle interior, powering headlights, etc.), or run the engine at a higher-than-idle (more efficient) power setting and use the excess

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power (over auxiliary loads) to recharge the storage device. When the vehicle control system can shut the engine off at idle, the drivetrain can be designed so that the drive motor also serves as the starter motor, allowing extremely rapid restart due to the motor's high starting torque.

- **Smaller ICE:** Since the storage device can take up a part of the load, the HEV's ICE can be down sized. The ICE may be sized for the continuous load and not for the very high short term acceleration load. This enables the ICE to operate at a higher fraction of its rated power, generally at higher fuel efficiency, during most of the driving.

There are counterbalancing factors reducing hybrids' energy advantage, including:

- **Potential for higher weight.** Although the fuel-driven energy source on a hybrid generally will be of lower power and weight than the engine in a conventional vehicle of similar performance, total hybrid weight is likely to be higher than the conventional vehicle it replaces because of the added weight of the storage device, electric motor(s), and other components. This depends, of course, on the storage mechanism chosen, the vehicle performance requirements, and so forth.
- **Electrical losses.** Although individual electric drivetrain components tend to be quite efficient for one-way energy flows, in many hybrid configurations, electricity flows back and forth through components in a way that leads to cascading losses. Further, some of the components may be forced to operate under conditions where they have reduced efficiency. For example, like ICEs, most electric motors have lower efficiency at the low-speed, low-load conditions often encountered in city driving. Without careful component selection and a control strategy that minimizes electric losses, much of the theoretical efficiency advantage often associated with an electric drivetrain can be lost.

CONCEPT OF HYBRID ELECTRIC DRIVE TRAINS

In Figure 2 the generic concept of a hybrid drivetrain and possible energy flow route is shown. The various possible ways of combining the power flow to meet the driving requirements are:

i. **powertrain 1 alone delivers its power of the load**

In gasoline (diesel)–IC engine (Energy source 1) and battery–electric motor (Energy source 2), Pattern 1 (powertrain 1 alone delivers power to the load) is the engine alone propelling mode. This may be used when the batteries are almost completely depleted,

and the engine has no remaining power to charge the batteries, or when the batteries have been fully charged, and the engine is able to supply sufficient power to meet vehicle power demands.

ii. powertrain 2 alone delivers its power to the load

Pattern 2 (powertrain 2 alone delivers power to the load) is the pure electric propelling mode, in which the engine is shut off. This pattern may be used for situations where the engine cannot operate effectively, such as at very low speed or in areas where emissions are strictly prohibited.

iii. both powertrain 1 and 2 deliver their power to load simultaneously.

Pattern 3 is the hybrid traction mode (both sources deliver power to load simultaneously) may be used when high power is needed, such as during sharp acceleration or steep hill climbing.

iv. powertrain 2 obtains power from the load (regenerative braking)

Pattern 4 is the regenerative braking mode, in which the kinetic or potential energy of the vehicle is recovered through the electric motor functioning as a generator. The recovered energy is then stored in the batteries and reused later.

v. powertrain 2 obtains power from powertrain 1

Pattern 5 Here the engine charges the batteries while the vehicle is at a standstill, coasting, or descending on a slight grade, in which no power goes into or comes from the load.

vi. powertrain 2 obtains power from powertrain 1 and load simultaneously

Pattern 6 is the mode in which both regenerating braking and the IC engine charge the batteries simultaneously.

vii. powertrain 1 delivers its power simultaneously to load and to powertrain 2

Pattern 7 is the mode in which the engine propels the vehicle and charges the batteries simultaneously.

viii. powertrain 1 delivers power to powertrain 2 and powertrain 2 delivers its power to the load

Pattern 8 is the mode in which the engine charges the batteries, and the batteries supply power to the load.

ix. powertrain 1 delivers its power to the load and load delivers power to powertrain 2.

Pattern 9 is the mode in which the power flows into the batteries from the heat engine through the vehicle mass. The typical configuration of this mode is that the two power trains are separately mounted on the front and rear axles of the vehicle.

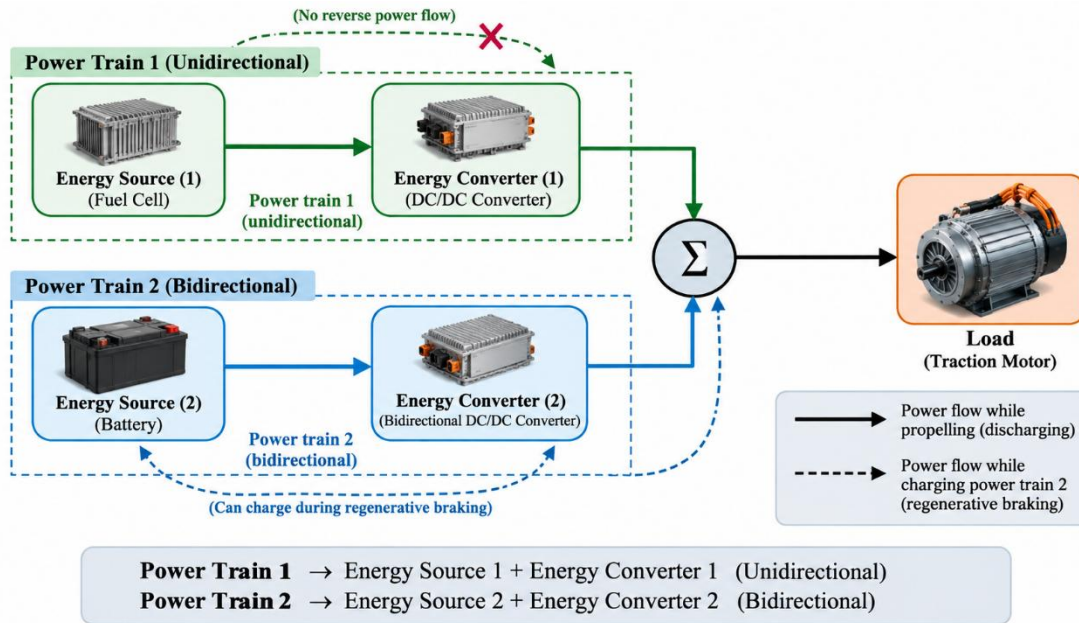


Figure 2: Generic Hybrid Drivetrain

The load power of a vehicle varies randomly in actual operation due to frequent acceleration, deceleration and climbing up and down the grades. The power requirement for a typical driving scenario is shown in **Figure 3**. The load power can be decomposed into two parts:

- steady power, i.e. the power with a constant value
- dynamic power, i.e. the power whose average value is zero

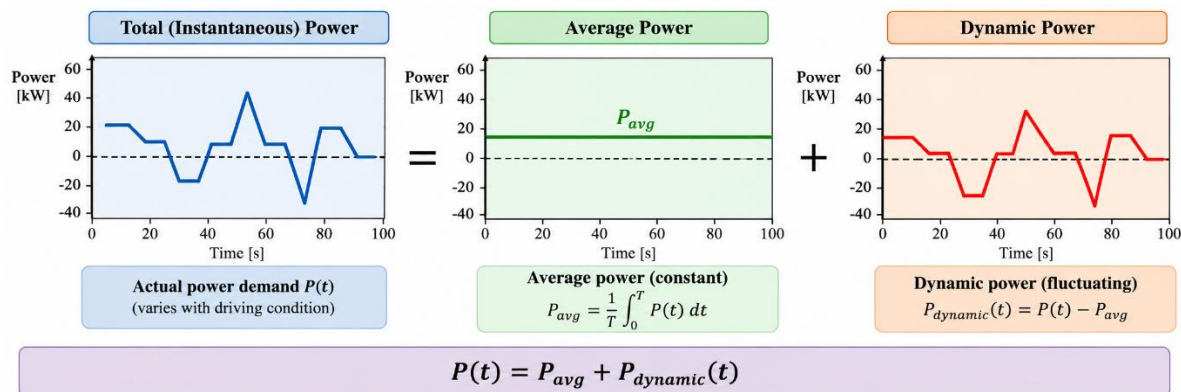


Figure 3: Load Power Decomposition

ARCHITECTURES OF HYBRID ELECTRIC DRIVE TRAINS

In HEV one powertrain favours steady state operation, such as an ICE or fuel cell. The other powertrain in the HEV is used to supply the dynamic power. The total energy output from the dynamic powertrain will be zero in the whole driving cycle. Generally, electric motors are used to meet the dynamic power demand. This hybrid drivetrain concept can be implemented by different configurations as follows:

- Series configuration
- Parallel configuration
- Series-parallel configuration
- Complex configuration

In **Figure 4** the functional block diagrams of the various HEV configurations is shown. From **Figure 4** it can be observed that the key feature of:

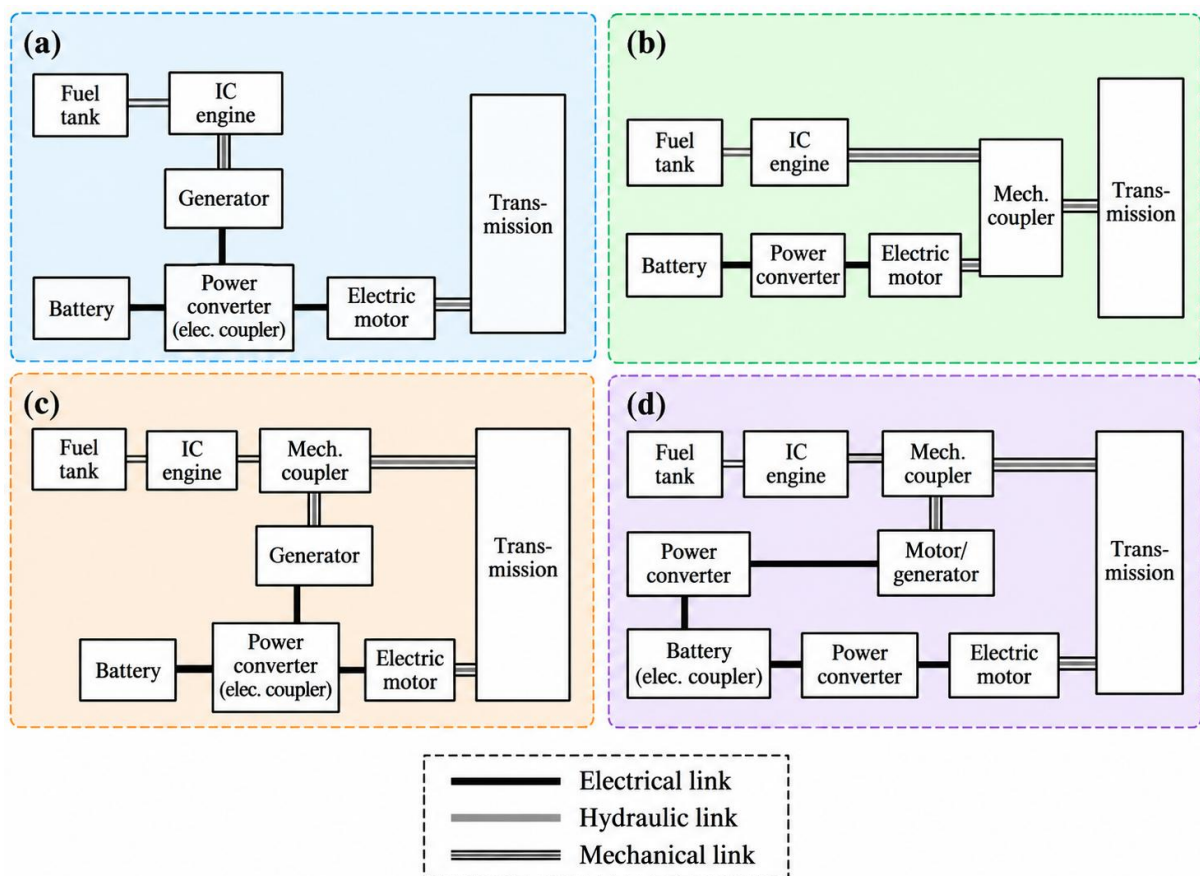


Figure 4: Classification of Hybrid EVs

Actually, in an HEV, there are two kinds of energy flowing in the drive train: one is mechanical energy and the other is electrical energy. Adding two powers together or splitting one power into two at the power merging point always occurs with the same power type, that is, electrical or mechanical, not electrical and mechanical. So perhaps a more accurate definition for HEV architecture may be to take the power coupling or decoupling features such as an electrical coupling drive train, a mechanical coupling drive train, and a mechanical–electrical coupling drive train.

Figure 4a functionally shows the architecture that is traditionally called a series hybrid drive train. The key feature of this configuration is that two electrical powers are added together in the power converter, which functions as an electric power coupler to control the power flows from the batteries and generator to the electric motor, or in the reverse direction, from the electric motor to the batteries. The fuel tank, the IC engine, and the generator constitute the primary energy supply and the batteries function as the energy bumper.

Figure 4b is the configuration that is traditionally called a parallel hybrid drive train. The key of this configuration is that two mechanical powers are added together in a mechanical coupler. The IC engine is the primary power plant, and the batteries and electric motor drive constitute the energy bumper. The power flows can be controlled only by the power plants—the engine and electric motor.

Figure 5.3c shows the configuration that is traditionally called a series parallel hybrid drive train. The distinguished feature of this configuration is the employment of two power couplers—mechanical and electrical. Actually, this configuration is the combination of series and parallel structures, possessing the major features of both and more plentiful operation modes than those of the series or parallel structure alone. On the other hand, it is relatively more complicated and may be of higher cost.

Figure 4d shows a configuration of the so-called complex hybrid, which has a similar structure to the series–parallel one. The only difference is that the electric coupling function is moved from the power converter to the batteries and one more power converter is added between the motor/generator and the batteries

Series Hybrid Electric DriveTrains (Electrical Coupling)

In case of series hybrid system (Figure 4a) the mechanical output is first converted into electricity using a generator. The converted electricity either charges the battery or can bypass the battery to propel the wheels via the motor and mechanical transmission. Conceptually, it is an ICE assisted Electric Vehicle (EV).

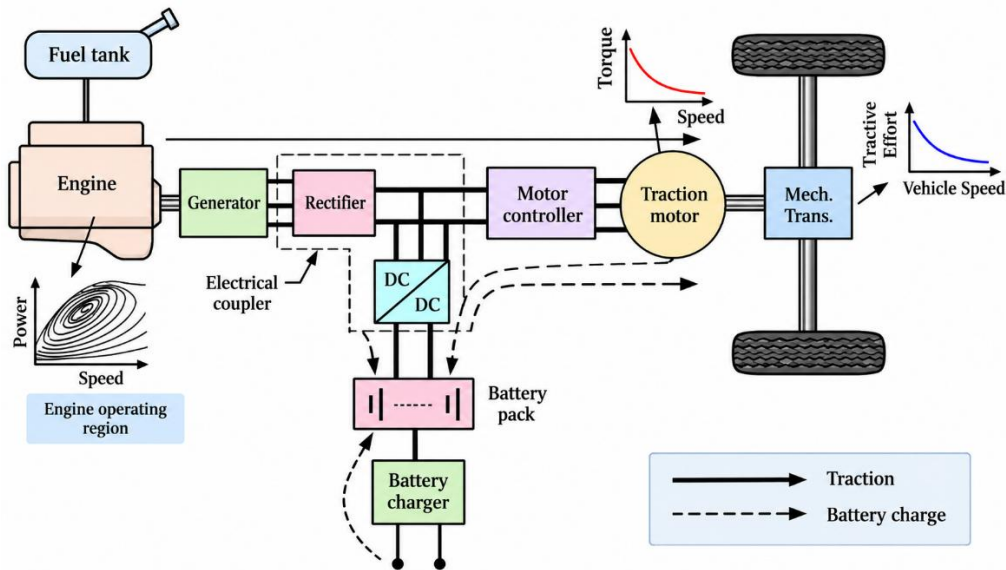


Figure 5: Configuration of a series hybrid electric drive train.

The advantages of series hybrid drivetrains are:

- mechanical decoupling between the ICE and driven wheels allows the IC engine operating at its very narrow optimal region as shown in Figure 5.
- nearly ideal torque-speed characteristics of electric motor make multigear transmission unnecessary.

However, a series hybrid drivetrain has the following disadvantages:

- the energy is converted twice (mechanical to electrical and then to mechanical) and this reduces the overall efficiency.
- Two electric machines are needed and a big traction motor is required because it is the only torque source of the driven wheels.

The series hybrid drivetrain is used in heavy commercial vehicles, military vehicles and buses. The reason is that large vehicles have enough space for the bulky engine/generator system.

Parallel Hybrid System:

The parallel HEV (Figure 4b) allows both ICE and electric motor (EM) to deliver power to drive the wheels. Since both the ICE and EM are coupled to the drive shaft of the wheels via two clutches, the propulsion power may be supplied by ICE alone, by EM only or by both ICE and EM. The EM can be used as a generator to charge the battery by regenerative braking or absorbing power from the ICE when its output is greater than that required to drive the wheels.

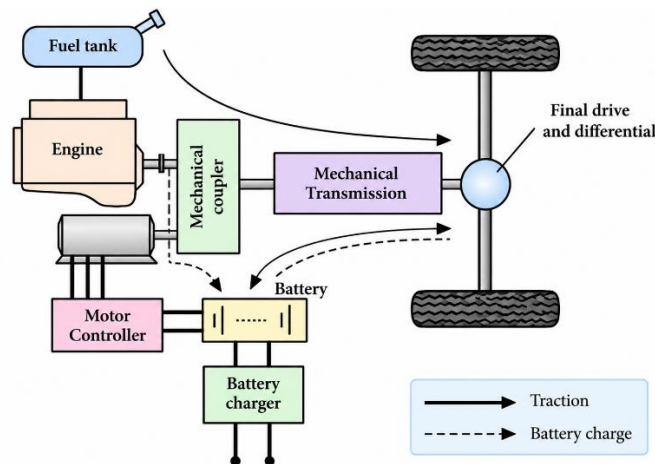


Figure 6: Configuration of a parallel hybrid electric drive train

The advantages of the parallel hybrid drivetrain are:

- both engine and electric motor directly supply torques to the driven wheels and no energy form conversion occurs, hence energy loss is less
- compactness due to no need of the generator and smaller traction motor.

The drawbacks of parallel hybrid drivetrains are:

- mechanical coupling between the engines and the driven wheels, thus the engine operating points cannot be fixed in a narrow speed region.
- The mechanical configuration and the control strategy are complex compared to series hybrid drivetrain.

Due to its compact characteristics, small vehicles use parallel configuration. Most passenger cars employ this configuration.

Series-Parallel System

In the series-parallel hybrid (Figure 4c), the configuration incorporates the features of both the series and parallel HEVs. However, this configuration needs an additional electric machine and a planetary gear unit making the control complex.

Complex Hybrid System

The complex hybrid system (Figure 4d) involves a complex configuration which cannot be classified into the above three kinds. The complex hybrid is similar to the series-parallel hybrid since the generator and electric motor is both electric machines. However, the key difference is due to the bi-directional power flow of the electric motor in complex hybrid and the unidirectional power flow of the generator in the series-parallel hybrid. The major disadvantage of complex hybrid is higher complexity.

POWER FLOW CONTROL

Due to the variations in HEV configurations, different power control strategies are necessary to regulate the power flow to or from different components. All the control strategies aim satisfy the following goals:

- maximum fuel efficiency
- minimum emissions
- minimum system costs
- good driving performance

The design of power control strategies for HEVs involves different considerations such as:

- ***Optimal ICE operating point:*** The optimal operating point on the torque-speed plane of the ICE can be based on maximization of fuel economy, the minimization of emissions or a compromise between fuel economy and emissions.
- ***Optimal ICE operating line:*** In case the ICE needs to deliver different power demands, the corresponding optimal operating points constitute an optimal operating line.
- ***Safe battery voltage:*** The battery voltage may be significantly altered during discharging, generator charging or regenerative charging. This battery voltage should not exceed the maximum voltage limit nor should it fall below the minimum voltage limit.

POWER FLOW CONTROL IN SERIES HYBRID

In the series hybrid system there are four operating modes based on the power flow:

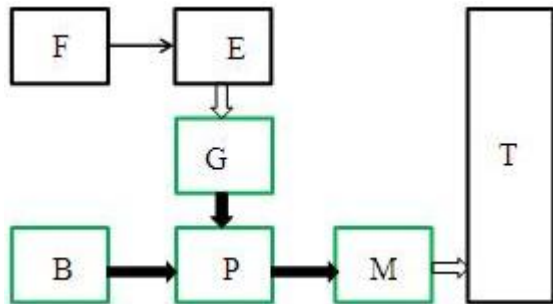


Figure 7a: Mode 1, normal driving or acceleration

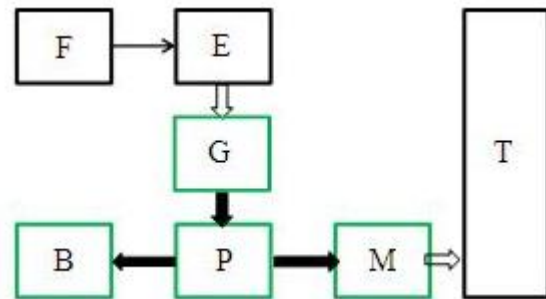


Figure 7b: Mode 2, light load

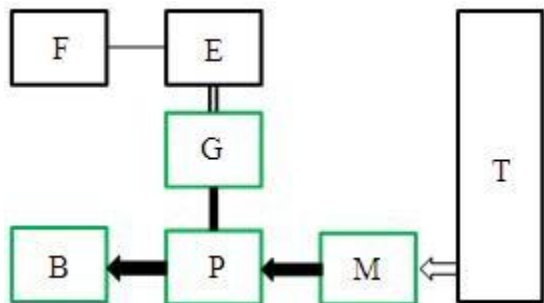


Figure 7c: Mode 3, braking or deceleration

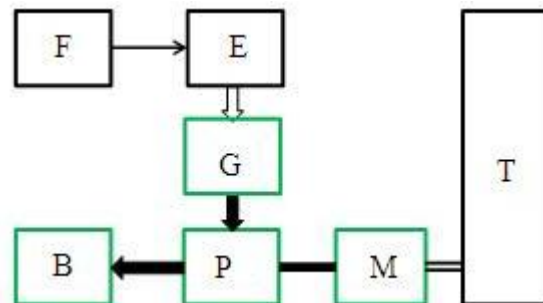


Figure 7d: Mode 4, vehicle at stop

B: Battery
E: ICE
F: Fuel tank
G: Generator
M: Motor
P: Power Converter

— Electrical link
— Hydraulic link
= Mechanical link

T: Transmission (including brakes, clutches and gears)

Figure 7: Power Flow Control in Series Hybrid

- **Mode 1:** During startup (Figure 7a), normal driving or acceleration of the series HEV, both the ICE and battery deliver electric energy to the power converter which then drives the electric motor and hence the wheels via transmission.
- **Mode 2:** At light load (Figure 7b), the ICE output is greater than that required to drive the wheels. Hence, a fraction of the generated electrical energy is used to charge the battery. The charging of the battery takes place till the battery capacity reaches a proper level.
- **Mode 3:** During braking or deceleration (Figure 7c), the electric motor acts as a generator, which converts the kinetic energy of the wheels into electricity and this, is used to charge the battery.

- **Mode 4:** The battery can also be charged by the ICE via the generator even when the vehicle comes to a complete stop (Figure 7d).

Power Flow Control in Parallel Hybrid

The parallel hybrid system has four modes of operation. These four modes of operation are

- **Mode 1:** During start up or full throttle acceleration (Figure 8a) ; both the ICE and the EM share the required power to propel the vehicle. Typically, the relative distribution between the ICE and electric motor is 80-20%.
- **Mode 2:** During normal driving (Figure 8b), the required traction power is supplied by the ICE only and the EM remains in off mode.
- **Mode 3:** During braking or deceleration (Figure 8c), the EM acts as a generator to charge the battery via the power converter.
- **Mode 4:** Under light load condition (Figure 8d), the traction power is delivered by the ICE and the ICE also charges the battery via the EM.

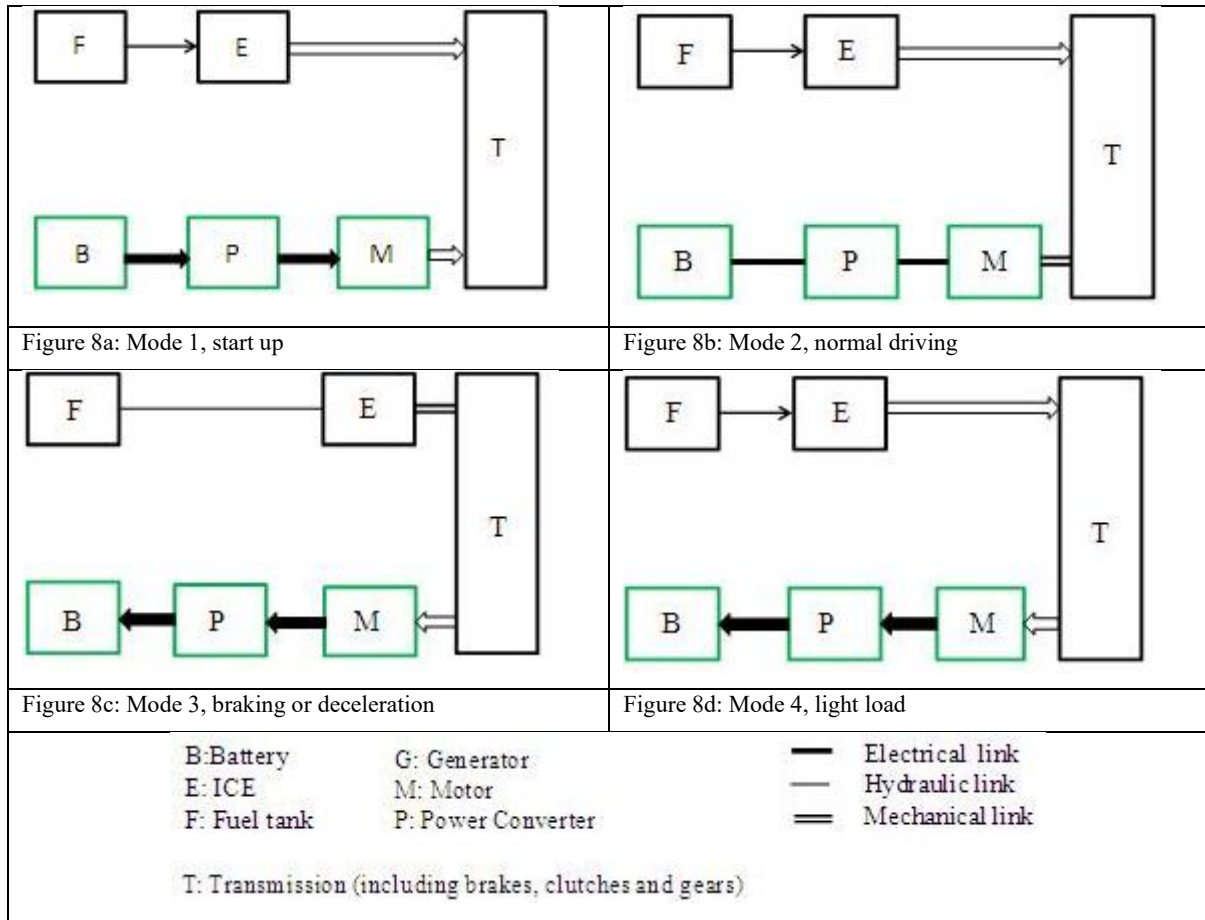


Figure 8: Power Flow Control in Parallel Vehicle

Power Flow Control in Series-Parallel Hybrid

The series-parallel hybrid system involves the features of series and parallel hybrid systems. Hence, a number of operation modes are feasible. Therefore, these hybrid systems are classified into two categories: **the ICE dominated** and **the EM dominated**.

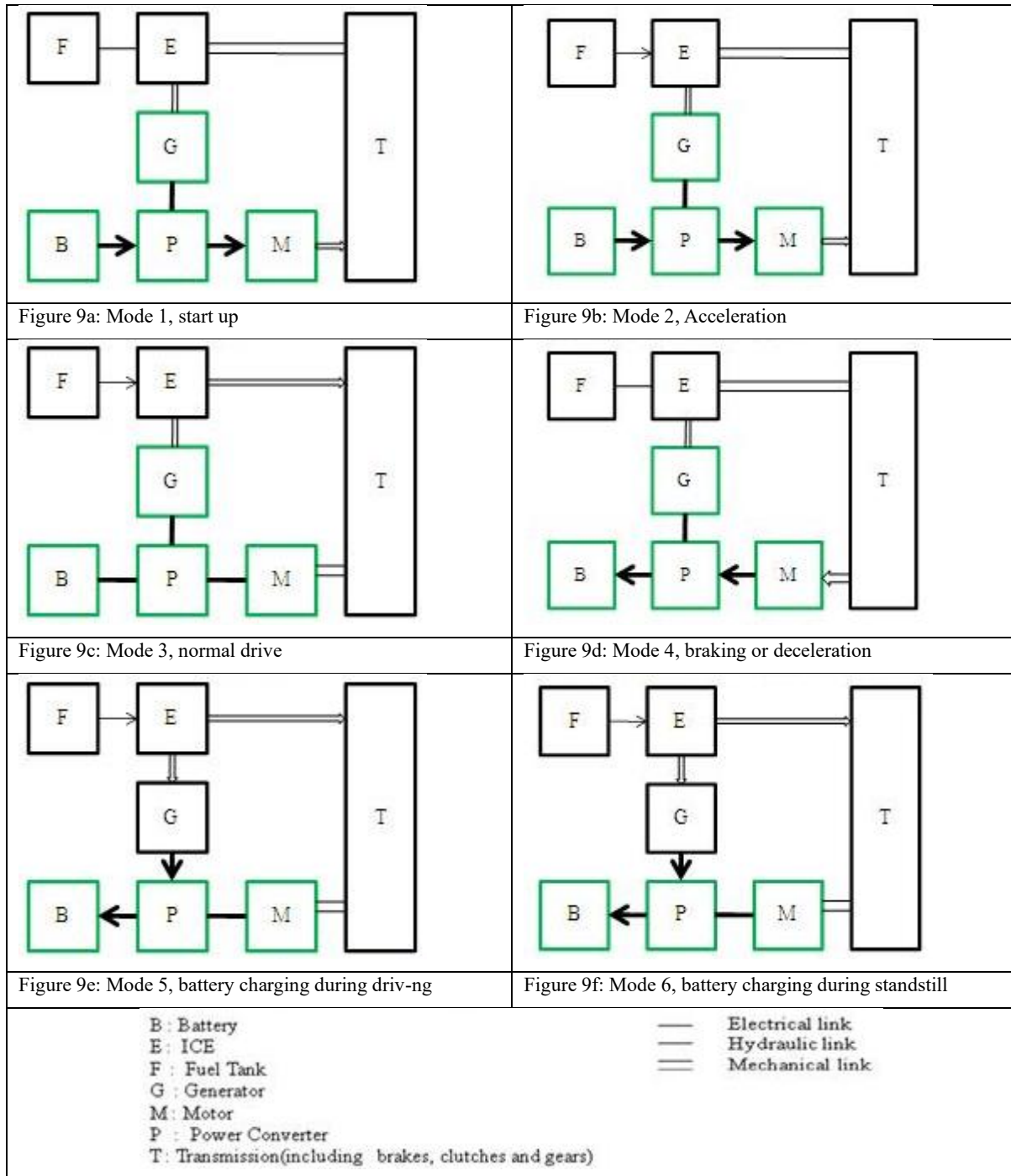


Figure 9: Power Flow Control of Series-Parallel Hybrid

The various operating modes of **ICE dominated** system are:

- **Mode 1:** At startup (Figure 9a) , the battery solely provides the necessary power to propel the vehicle and the ICE remains in off mode.

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- **Mode 2:** During full throttle acceleration (Figure 9b), both the ICE and the EM share the required traction power.
- **Mode 3:** During normal driving (Figure 9c), the required traction power is provided by the ICE only and the EM remains in the off state.
- **Mode 4:** During normal braking or deceleration (Figure 9d), the EM acts as a generator to charge the battery.
- **Mode 5:** To charge the battery during driving (Figure 9e), the ICE delivers the required traction power and also charges the battery. In this mode the EM acts as a generator.
- **Mode 6:** When the vehicle is at standstill (Figure 9f), the ICE can deliver power to charge the battery via the EM

The operating modes of **EM dominated** system are:

- **Mode 1:** During startup (Figure 10a), the EM provides the traction power and the ICE remains in the off state.
- **Mode 2:** During full throttle (Figure 10b), both the ICE and EM provide the traction power.
- **Mode 3:** During normal driving (Figure 10c), both the ICE and EM provide the traction power.
- **Mode 4:** During braking or deceleration (Figure 10d), the EM acts as a generator to charge the battery.
- **Mode 5:** To charge the battery during driving (Figure 10e), the ICE delivers the required traction power and also charges the battery. The EM acts as a generator.
- **Mode 6:** When the vehicle is at standstill (Figure 10f), the ICE can deliver power to charge the battery via the EM

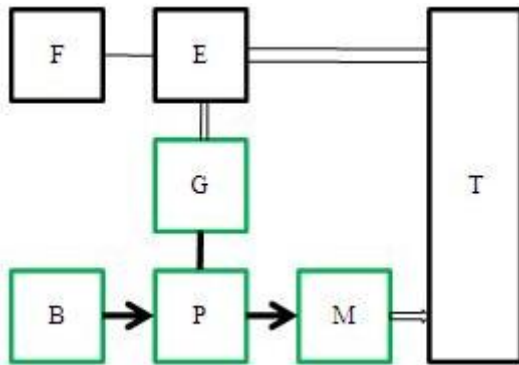


Figure 10a: Mode 1 , start up

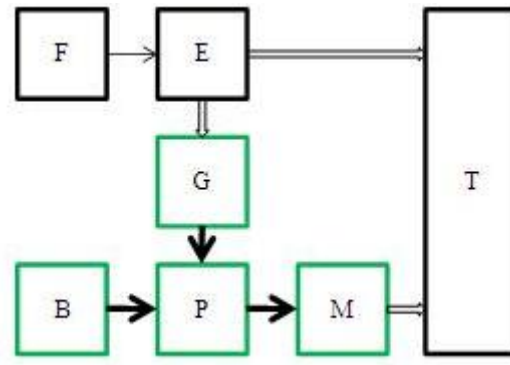


Figure 10b: Mode 2, acceleration

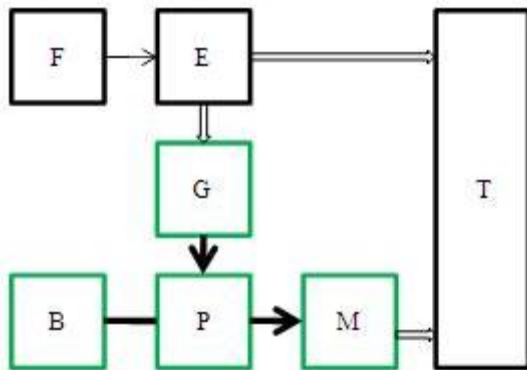


Figure 10c: Mode 3, normal Drive

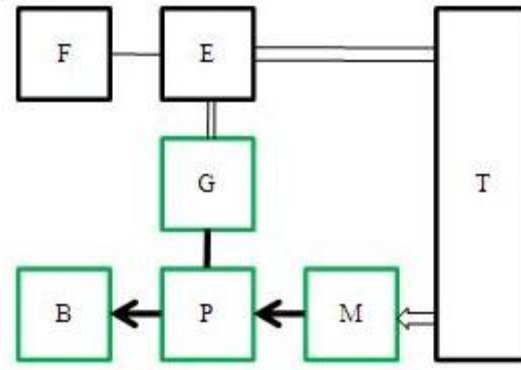


Figure 10d: Mode 4 , Braking or deceleration

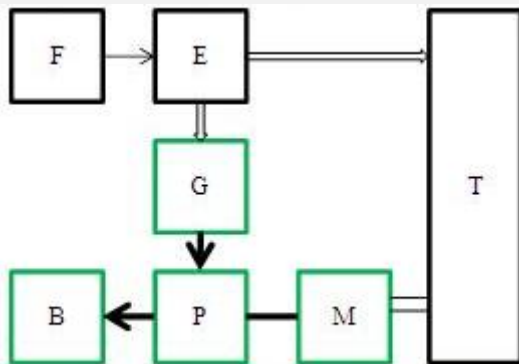


Figure 10e: Mode 5, battery charging during driving

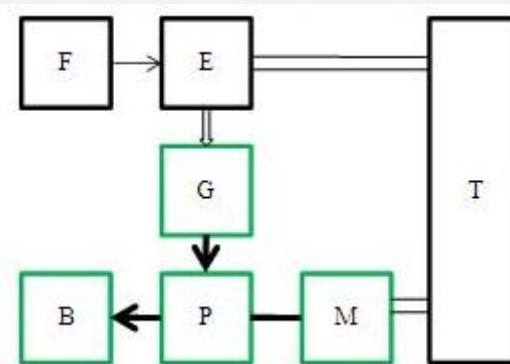


Figure 10f: Mode 6, battery charging during standstill

B : Battery
E : ICE
F : Fuel Tank
G : Generator
M : Motor
P : Power Converter
T : Transmission(including brakes, clutches and gears)

— Electrical link
— Hydraulic link
— Mechanical link

Figure 10: : Power Flow Control of Series-Parallel Hybrid

Power Flow Control Complex Hybrid Control

The complex hybrid vehicle configurations are of two types:

- Front hybrid rear electric
- Front electric and rear hybrid

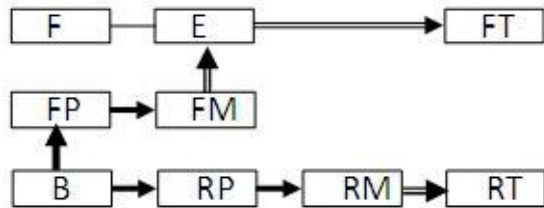


Figure 11a: Mode 1, Startup

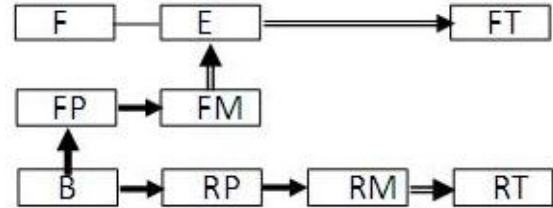


Figure 11b: Mode 2, full throttle acceleration

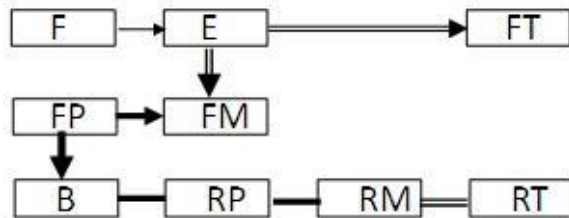


Figure 11c: Mode 3, vehicle properl and battery charging

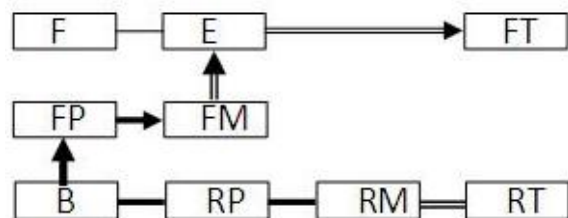


Figure 11d: Mode 4, light load

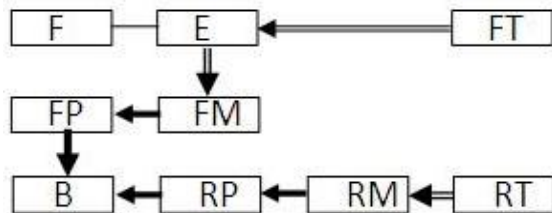


Figure 11e: Mode 5, Braking or decelration

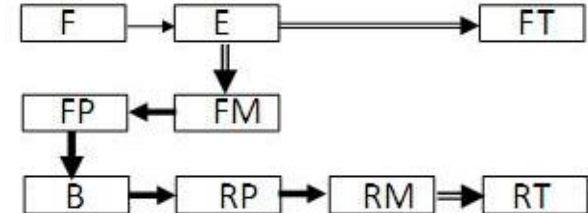


Figure 11f: Mode 1, axil balancing

B: Battery FM: Front motor FP: Front power converter FT: Front axel transmission E: ICE F: Fuel tank
RM: Rear motor RP: Rear power converter RT: Rear axel transmission

— Electrical link
— Hydraulic link
— Mechanical link

Figure 11 all the six modes of operation of front electric and rear hybrid are shown.

Both the configurations have six modes of operation:

- **Mode 1:** During startup (Figure 11a), the required traction power is delivered by the EMs and the engine is in off mode.

- **Mode 2:** During full throttle acceleration (Figure 11b), both the ICE and the front wheel EM deliver the power to the front wheel and the second EM delivers power to the rear wheel.
- **Mode 3:** During normal driving (Figure 11c), the ICE delivers power to propel the front wheel and to drive the first EM as a generator to charge the battery.
- **Mode 4:** During driving at light load (Figure 11d) first EM delivers the required traction power to the front wheel. The second EM and the ICE are in off state.
- **Mode 5:** During braking or deceleration (Figure 11e), both the front and rear wheel EMs act as generators to simultaneously charge the battery.
- **Mode 6:** A unique operating mode of complex hybrid system is **axial balancing**. In this mode (Figure 11f) if the front wheel slips, the front EM works as a generator to absorb the change of ICE power. Through the battery, this power difference is then used to drive the rear wheels to achieve the axle balancing.

In **Figures 12a-f** all the six modes of operation of front electric and rear hybrid are shown.

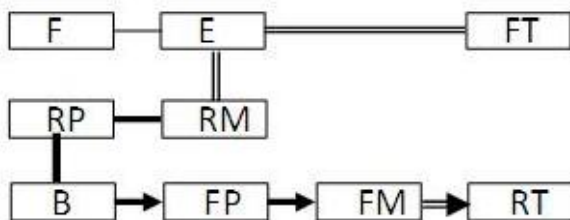


Figure 12a: Mode 1, Startup

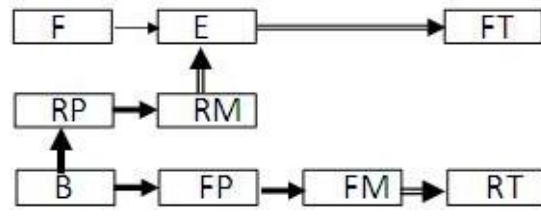


Figure 12b: Mode 2, full throttle acceleration

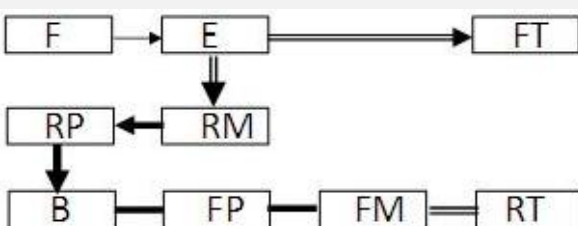


Figure 12c: Mode 3, vehicle properl and battery charging

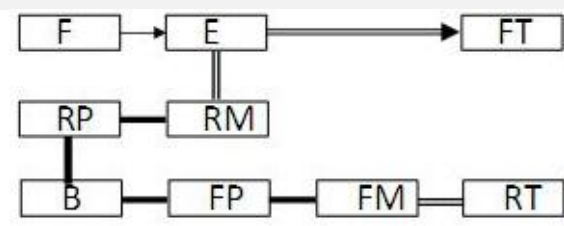


Figure 12d: Mode 4, light load

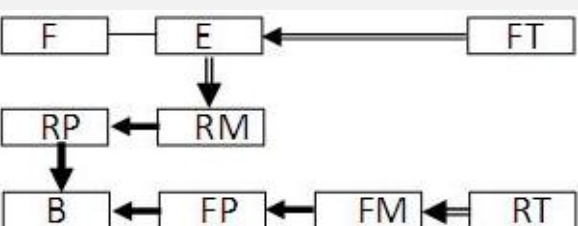


Figure 12e: Mode 5, Braking or deceleration

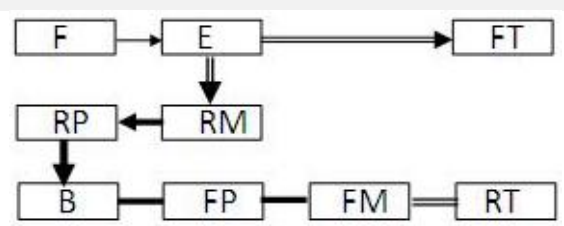


Figure 12f: Mode 1, axil balancing

ELECTRIC VEHICLES (EEN5003)

DR. VENU SANGWAN



B: Battery RM: Rear motor FP: Front power converter FT: Front axel transmission E: ICE F: Fuel tank
RM: Rear motor RP: Rear power converter RT: Rear axel transmission

— Electrical link
— Hydraulic link
= Mechanical link